Lecture 10 — The QR Decomposition and Orthogonality

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 •
Tuesday, October 13, 2026 (Monday schedule)

After this lecture you should be able to

- Compute inner products and Euclidean lengths in $\mathbb{R}^n$, and test two vectors for orthogonality.
- Explain why an orthonormal basis makes $\boldsymbol{U}^{-1} = \boldsymbol{U}^{\mathsf{T}}$, and expand a vector in such a basis with a single transpose.
- Run the Gram–Schmidt process to build an orthonormal basis for the range of a matrix.
- Read $\boldsymbol{A} = \boldsymbol{QR}$ as a record of that process, and use it to solve $\boldsymbol{Ax} = \boldsymbol{b}$ by back-substitution.

Where we are

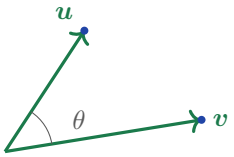
Elimination gave us $\boldsymbol{A} = \boldsymbol{LU}$ and a way to solve square systems. Today we build the second great decomposition of the course. Its raw material is the dot product, its engine is Gram–Schmidt, and its payoff is that inverting an orthogonal matrix costs nothing at all: you transpose it.

1 Orthogonal vectors and orthonormal bases

The dot product / inner product in $\mathbb{R}^n$

Take $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$, written as columns

$$\boldsymbol{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \qquad \boldsymbol{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \qquad \longleftarrow \text{ representation in the Cartesian basis.}$$



The dot product has a geometric definition and an algebraic one, and they agree:

$$\boldsymbol{u} \cdot \boldsymbol{v} = \|\boldsymbol{u}\| \, \|\boldsymbol{v}\| \cos\theta = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{v} \quad \text{(matrix multiplication)}.$$

In particular, the Euclidean length squared of $\boldsymbol{u}$ is

$$\|\boldsymbol{u}\|^2 = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{u} = u_1^2 + \cdots + u_n^2,$$

which is Pythagoras' theorem in n dimensions.

Two vectors $\boldsymbol{u}, \boldsymbol{v}$ are **orthogonal** if

$$\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u}^{\mathsf{T}}\boldsymbol{v} = 0 \qquad (\cos\theta = 0 \implies \theta = \pm\tfrac{\pi}{2}).$$

A vector $\boldsymbol{u}$ (or $\boldsymbol{v}$) is **normalised** if $\|\boldsymbol{u}\| = 1$, equivalently $\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u} = 1$.

Example 1.1 (Lengths, angles and orthogonality)**. Given.** Three vectors in $\mathbb{R}^3$, written in a row to save space:

$$\boldsymbol{u} = (1, 1, 0), \qquad \boldsymbol{v} = (1, 0, 1), \qquad \boldsymbol{w} = (-1, 1, 1).$$

Find. (a) What are the lengths of the three vectors, and the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$? (b) Which pairs are orthogonal? (c) What is the unit vector in the direction of $\boldsymbol{w}$? (d) Is $\|\boldsymbol{u}+\boldsymbol{w}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{w}\|^2$? Is $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = \|\boldsymbol{u}\|^2 + \|\boldsymbol{v}\|^2$?

Answer.

(a) $\|\boldsymbol{u}\|^2 = \boldsymbol{u}^\mathsf{T}\boldsymbol{u} = 1+1+0 = 2$, so $\|\boldsymbol{u}\| = \sqrt{2}$. In the same way $\|\boldsymbol{v}\| = \sqrt{2}$ and $\|\boldsymbol{w}\| = \sqrt{3}$. For the angle, $\boldsymbol{u}^\mathsf{T}\boldsymbol{v} = 1\cdot 1 + 1\cdot 0 + 0\cdot 1 = 1$, and

$$\cos\theta = \frac{\boldsymbol{u}^\mathsf{T}\boldsymbol{v}}{\|\boldsymbol{u}\|\,\|\boldsymbol{v}\|} = \frac{1}{\sqrt{2}\,\sqrt{2}} = \frac{1}{2}, \qquad \theta = \frac{\pi}{3} \;\; (60^\circ).$$

(b) $\boldsymbol{u}^\mathsf{T}\boldsymbol{w} = -1+1+0 = 0$ and $\boldsymbol{v}^\mathsf{T}\boldsymbol{w} = -1+0+1 = 0$. So $\boldsymbol{w}$ is orthogonal to both $\boldsymbol{u}$ and $\boldsymbol{v}$, and therefore to every combination $c\,\boldsymbol{u} + d\,\boldsymbol{v}$: it is perpendicular to the plane that $\boldsymbol{u}$ and $\boldsymbol{v}$ span. The pair $\boldsymbol{u}, \boldsymbol{v}$ is not orthogonal, since $\boldsymbol{u}^\mathsf{T}\boldsymbol{v} = 1$.

(c) Divide by the length: $\boldsymbol{w}/\|\boldsymbol{w}\| = (-1,1,1)/\sqrt{3}$. Its length squared is $(1+1+1)/3 = 1$.

(d) Multiplying out,

$$\|\boldsymbol{u}+\boldsymbol{v}\|^2 = (\boldsymbol{u}+\boldsymbol{v})^\mathsf{T}(\boldsymbol{u}+\boldsymbol{v}) = \|\boldsymbol{u}\|^2 + 2\,\boldsymbol{u}^\mathsf{T}\boldsymbol{v} + \|\boldsymbol{v}\|^2,$$

and the same with $\boldsymbol{w}$ in place of $\boldsymbol{v}$. So Pythagoras holds exactly when the cross term is zero. For $\boldsymbol{u}, \boldsymbol{w}$ it is: $\boldsymbol{u}+\boldsymbol{w} = (0,2,1)$ and $\|\boldsymbol{u}+\boldsymbol{w}\|^2 = 5 = 2+3$. For $\boldsymbol{u}, \boldsymbol{v}$ it is not: $\boldsymbol{u}+\boldsymbol{v} = (2,1,1)$ and $\|\boldsymbol{u}+\boldsymbol{v}\|^2 = 6$, which is $2+2$ plus the cross term $2\,\boldsymbol{u}^\mathsf{T}\boldsymbol{v} = 2$.

Bases

A basis $U = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ is a set of linearly independent vectors that span a vector space.

Standard example: the Cartesian basis in $\mathbb{R}^n$.

$$\boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \boldsymbol{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots \quad \boldsymbol{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

A "default" basis. Put together, they give the identity matrix $\boldsymbol{I}_n = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$.

Generally, n linearly independent vectors $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ form a basis for $\mathbb{R}^n$, and we collect them as the columns of a matrix

$$\boldsymbol{U} = \begin{pmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_n \\ | & & | \end{pmatrix}.$$

One interpretation of the linear system $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$: find coefficients $x_1, \ldots, x_n$ to represent the vector $\boldsymbol{b}$ using the U basis,

$$\boldsymbol{b} = x_1\boldsymbol{u}_1 + \cdots + x_n\boldsymbol{u}_n.$$

Formally, $\boldsymbol{x} = \boldsymbol{U}^{-1}\boldsymbol{b}$, and this makes sense because $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ linearly independent $\iff$ $\boldsymbol{U}$ is invertible.

Key idea

Solving $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$ is a change of coordinates: it re-expresses the vector $\boldsymbol{b}$, which arrived in Cartesian coordinates, in the basis U. The difficulty is entirely in $\boldsymbol{U}^{-1}$ — and calculating $\boldsymbol{U}^{-1}$ for a general invertible $\boldsymbol{U}$ is costly.

Example 1.2 (Basis or not?)**. Given.** Three sets of vectors in $\mathbb{R}^2$:

$$S_1 = \{(2,1),\,(1,1)\}, \qquad S_2 = \{(1,2),\,(2,4)\}, \qquad S_3 = \{(1,0),\,(0,1),\,(1,1)\}.$$

Find. Which of them is a basis for $\mathbb{R}^2$?

Answer. A basis has to do two things: span the space, and be linearly independent, so that every vector has exactly one set of coefficients.

(S_1) Yes. Put the two vectors into the columns of $\boldsymbol{U} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Then $\det \boldsymbol{U} = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, so $\boldsymbol{U}$ is invertible: the columns are independent and span $\mathbb{R}^2$.

(S_2) No. The second vector is twice the first, $(2,4) = 2\,(1,2)$, so every combination is a multiple of $(1,2)$: the two vectors span only a line. The determinant $1 \cdot 4 - 2 \cdot 2$ is zero.

(S_3) No, although these three vectors do span $\mathbb{R}^2$. They are dependent, since $(1,1) = (1,0) + (0,1)$, so the coefficients are not unique:

$$(1,1) = 1\,(1,0) + 1\,(0,1) + 0\,(1,1) = 0\,(1,0) + 0\,(0,1) + 1\,(1,1).$$

A basis of $\mathbb{R}^n$ always has exactly n vectors; here there is one too many.

Example 1.3 (Coordinates in a basis that is not orthonormal)**. Given.** The basis $\boldsymbol{u}_1 = (2,1)$, $\boldsymbol{u}_2 = (1,1)$ (the set S_1 of Example 1.2) and the vector $\boldsymbol{b} = (3,1)$.
Find. The coefficients in $\boldsymbol{b} = x_1\boldsymbol{u}_1 + x_2\boldsymbol{u}_2$.
Answer. This is the linear system $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$:

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Subtracting $\frac{1}{2}$ of row 1 from row 2 leaves $\frac{1}{2}x_2 = -\frac{1}{2}$, so $x_2 = -1$, and then $2x_1 - 1 = 3$ gives $x_1 = 2$. So $\boldsymbol{x} = (2,-1)$, and indeed $2\,(2,1) - (1,1) = (3,1)$.

The dot products $\boldsymbol{u}_1^{\mathsf{T}}\boldsymbol{b} = 7$ and $\boldsymbol{u}_2^{\mathsf{T}}\boldsymbol{b} = 4$ are not the coefficients. When the basis vectors are not orthogonal there is no such shortcut: we solve a system, or multiply by $\boldsymbol{U}^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. Example 2.2 finds the coefficients of the same $\boldsymbol{b}$ in an orthonormal basis.

A very nice type of basis: orthonormal bases

An **orthonormal basis** is one whose vectors are mutually orthogonal and each of unit length:

$$\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{u}_j = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

The Cartesian basis is an orthonormal basis. Another example, for $\mathbb{R}^2$:

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

Example 1.4 (Checking that a basis is orthonormal)**. Given.** Three vectors in $\mathbb{R}^3$:

$$\boldsymbol{q}_1 = \tfrac{1}{3}(2,2,-1), \qquad \boldsymbol{q}_2 = \tfrac{1}{3}(-1,2,2), \qquad \boldsymbol{q}_3 = \tfrac{1}{3}(2,-1,2).$$

Find. Is $\{\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3\}$ an orthonormal basis of $\mathbb{R}^3$?

Answer. Yes. It saves work to keep the factor $\frac{1}{3}$ aside and check the integer vectors $(2,2,-1)$, $(-1,2,2)$, $(2,-1,2)$ first. Each has length squared $1+4+4=9$, so each $\boldsymbol{q}_i$ has length $3/3=1$. Their dot products are

$$\begin{aligned} (2,2,-1)\cdot(-1,2,2) &= -2+4-2 = 0, \\ (2,2,-1)\cdot(2,-1,2) &= 4-2-2 = 0, \\ (-1,2,2)\cdot(2,-1,2) &= -2-2+4 = 0. \end{aligned}$$

So the three vectors are orthonormal. Orthonormal vectors are always independent: if $c_1\boldsymbol{q}_1 + c_2\boldsymbol{q}_2 + c_3\boldsymbol{q}_3 = \boldsymbol{0}$, take the dot product with $\boldsymbol{q}_1$. Every term but the first is zero, and the first is $c_1\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_1 = c_1$, so $c_1 = 0$; in the same way $c_2 = c_3 = 0$. Three independent vectors in $\mathbb{R}^3$ form a basis.

2 Orthogonal matrices

Take an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$. Then the matrix

$$\boldsymbol{U} = \begin{pmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_n \\ | & & | \end{pmatrix}$$

is called an **orthogonal matrix**, and it has the defining property

$$\boldsymbol{U}^{-1} = \boldsymbol{U}^{\mathsf{T}}.$$

Check it. Multiply $\boldsymbol{U}^{\mathsf{T}}$ (whose *rows* are the $\boldsymbol{u}_i^{\mathsf{T}}$) by $\boldsymbol{U}$ (whose *columns* are the $\boldsymbol{u}_j$):

$$\boldsymbol{U}^{\mathsf{T}}\boldsymbol{U} = \begin{pmatrix} — \ \boldsymbol{u}_1^{\mathsf{T}} \ — \\ \vdots \\ — \ \boldsymbol{u}_n^{\mathsf{T}} \ — \end{pmatrix} \begin{pmatrix} | & & | \\ \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_n \\ | & & | \end{pmatrix} = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix} = \boldsymbol{I}.$$

The (i,j) entry of the product is exactly $\boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{u}_j$, which is 1 on the diagonal and 0 off it — the orthonormality condition, read as a matrix identity.

So it is easy to calculate coefficients for an orthonormal basis:

$$\boldsymbol{x} = \boldsymbol{U}^{-1}\boldsymbol{b} = \boldsymbol{U}^{\mathsf{T}}\boldsymbol{b},$$

just a flip of the original $\boldsymbol{U}$. Componentwise, $x_i = \boldsymbol{u}_i^{\mathsf{T}}\boldsymbol{b}$: each coefficient is a single dot product.

Key idea

An orthonormal basis makes the hardest step free. For a general basis, changing coordinates costs a matrix inversion; for an orthonormal basis it costs a transpose. Every use of QR in this course is ultimately cashing in on that one fact.

"Orthogonal matrix" means *ortho normal* columns

The name is a historical accident. A matrix with merely orthogonal columns does not satisfy $\boldsymbol{U}^{\mathsf{T}}\boldsymbol{U} = \boldsymbol{I}$ — the diagonal entries come out $\|\boldsymbol{u}_i\|^2$ instead of 1. The normalisation is not decoration; without it $\boldsymbol{U}^{\mathsf{T}}$ is not the inverse.

Example 2.1 (Which of these are orthogonal matrices?)**. Given.** Four matrices:

$$\boldsymbol{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \boldsymbol{P} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \boldsymbol{W} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \boldsymbol{G} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Find. Which of them are orthogonal matrices? What is the inverse of each one that is?

Answer.

(a) $\boldsymbol{R}_\theta$, the rotation by the angle θ, is orthogonal:

$$\boldsymbol{R}_\theta^{\mathsf{T}}\boldsymbol{R}_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{pmatrix} = \boldsymbol{I}.$$

Its inverse $\boldsymbol{R}_\theta^{\mathsf{T}}$ is $\boldsymbol{R}_{-\theta}$, since $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$: to undo a rotation by θ, rotate back by θ.

(b) $\boldsymbol{P}$ is orthogonal. Its columns are $\boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_1$, the Cartesian basis in a different order, so $\boldsymbol{P}^{\mathsf{T}}\boldsymbol{P} = \boldsymbol{I}$ and $\boldsymbol{P}^{-1} = \boldsymbol{P}^{\mathsf{T}}$. The matrix $\boldsymbol{P}$ moves the entries of a vector, $\boldsymbol{P}(x_1, x_2, x_3) = (x_3, x_1, x_2)$, and $\boldsymbol{P}^{\mathsf{T}}$ moves them back. A matrix with a single 1 in each row and each column and zeros elsewhere is a permutation matrix, and every permutation matrix is orthogonal for the same reason.

(c) $\boldsymbol{W}$ is not. Its columns $(1, 1)$ and $(1, -1)$ are orthogonal but have length $\sqrt{2}$, and $\boldsymbol{W}^{\mathsf{T}}\boldsymbol{W} = 2\boldsymbol{I}$. Dividing by $\sqrt{2}$ repairs this: the columns of $\boldsymbol{W}/\sqrt{2}$ are the orthonormal basis of $\mathbb{R}^2$ in Section 1.

(d) $\boldsymbol{G}$ is not. Its columns $(1, 0)$ and $(1, 1)$ have dot product 1, and $\boldsymbol{G}^{\mathsf{T}}\boldsymbol{G} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \neq \boldsymbol{I}$.

Example 2.2 (Coordinates by dot products)**. Given.** (a) The orthonormal basis $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$, $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$ of $\mathbb{R}^2$ from Section 1, and $\boldsymbol{b} = (3, 1)$. (b) The orthonormal basis $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ of Example 1.4, collected in the orthogonal matrix

$$\boldsymbol{Q} = \frac{1}{3}\begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix}, \qquad \text{and} \qquad \boldsymbol{b} = (1, 2, 3).$$

Find. In each case, the coefficients in $\boldsymbol{b} = x_1\boldsymbol{q}_1 + x_2\boldsymbol{q}_2 + \cdots$.

Answer. No elimination is needed: $\boldsymbol{x} = \boldsymbol{Q}^{-1}\boldsymbol{b} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b}$, one dot product per coefficient.

(a) $x_1 = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{b} = (3 + 1)/\sqrt{2} = 2\sqrt{2}$ and $x_2 = \boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{b} = (3 - 1)/\sqrt{2} = \sqrt{2}$. Check: $2\sqrt{2}\,\boldsymbol{q}_1 + \sqrt{2}\,\boldsymbol{q}_2 = 2(1, 1) + (1, -1) = (3, 1)$. Compare Example 1.3, where the same $\boldsymbol{b}$, in a basis that is not orthogonal, needed elimination.

(b) $x_1 = \boldsymbol{q}_1^\mathsf{T}\boldsymbol{b} = (2+4-3)/3 = 1$, $x_2 = \boldsymbol{q}_2^\mathsf{T}\boldsymbol{b} = (-1+4+6)/3 = 3$ and $x_3 = \boldsymbol{q}_3^\mathsf{T}\boldsymbol{b} = (2-2+6)/3 = 2$. Check:

$$\boldsymbol{q}_1 + 3\,\boldsymbol{q}_2 + 2\,\boldsymbol{q}_3 = \tfrac{1}{3}\big[(2,2,-1) + (-3,6,6) + (4,-2,4)\big] = \tfrac{1}{3}\,(3,6,9) = (1,2,3).$$

Example 2.3 (Orthogonal columns that are not unit vectors)**. Given.** The matrix $\boldsymbol{W}$ of Example 2.1, whose columns $\boldsymbol{u}_1 = (1,1)$ and $\boldsymbol{u}_2 = (1,-1)$ are orthogonal with length $\sqrt{2}$, and $\boldsymbol{b} = (3,1)$. **Find.** (a) $\boldsymbol{W}^\mathsf{T}\boldsymbol{W}$ and $\boldsymbol{W}^{-1}$. (b) The coefficients in $\boldsymbol{b} = x_1\boldsymbol{u}_1 + x_2\boldsymbol{u}_2$.

Answer.

(a) $\boldsymbol{W}^\mathsf{T}\boldsymbol{W} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right) = 2\boldsymbol{I}$. The diagonal holds $\|\boldsymbol{u}_1\|^2 = \|\boldsymbol{u}_2\|^2 = 2$, the point of the warning box above, so $\boldsymbol{W}^\mathsf{T}$ is not the inverse; $\boldsymbol{W}^{-1} = \frac{1}{2}\boldsymbol{W}^\mathsf{T}$ is.

(b) $\boldsymbol{x} = \frac{1}{2}\boldsymbol{W}^\mathsf{T}\boldsymbol{b} = \frac{1}{2}\,(3+1,\ 3-1) = (2,1)$. Check: $2\,(1,1) + (1,-1) = (3,1)$. In general, when the columns are orthogonal but not unit vectors, each coefficient is a dot product divided by a length squared, $x_i = \boldsymbol{u}_i^\mathsf{T}\boldsymbol{b}\,/\,\boldsymbol{u}_i^\mathsf{T}\boldsymbol{u}_i$. Here each $\boldsymbol{u}_i$ is $\sqrt{2}$ times the $\boldsymbol{q}_i$ of Example 2.2(a), so its coefficient is $\sqrt{2}$ times smaller.

3 The range of a general rectangular matrix

Consider a general rectangular $\boldsymbol{A} \in \mathbb{R}^{m\times n}$. Suppose for the moment that $m > n$, and that $\boldsymbol{A}$ has full column rank — that is, $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ are linearly independent:

$$\boldsymbol{A} = \begin{pmatrix} | & & | \\ \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n \\ | & & | \end{pmatrix}.$$

Then $\boldsymbol{A}\boldsymbol{x}$, with $\boldsymbol{x} \in \mathbb{R}^n$, is

$$\boldsymbol{A}\boldsymbol{x} = \sum_i x_i \boldsymbol{a}_i,$$

a linear combination of the columns $\boldsymbol{a}_i$ of $\boldsymbol{A}$. Collecting all of them,

$$\text{Range}(\boldsymbol{A}) = \Big\{\textstyle\sum_i x_i\boldsymbol{a}_i\ ,\ \forall\,\boldsymbol{x} \in \mathbb{R}^n\Big\}: \quad \text{all possible linear combinations of the columns of } \boldsymbol{A}.$$

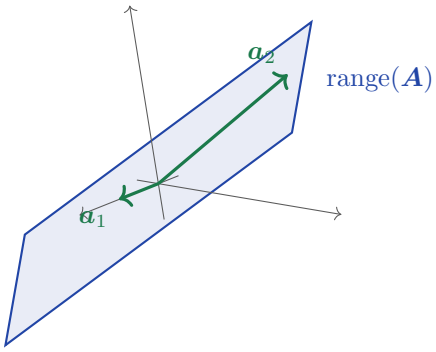
Range($\boldsymbol{A}$) is a lower dimensional subspace of $\mathbb{R}^m$, and $\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$ is a basis of Range($\boldsymbol{A}$).

Given what we know about orthonormal bases, it is natural to consider finding an *orthonormal* basis of Range($\boldsymbol{A}$).

Example 3.1. Take

$$\boldsymbol{A} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \text{range}(\boldsymbol{A}) = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix},$$

with columns $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$.



$\boldsymbol{A}$ is a 3×2 matrix: for 2-dimensional input vectors $\boldsymbol{x}$, $\boldsymbol{Ax}$ is a 3-dimensional vector. range($\boldsymbol{A}$) is the set of all possible $\boldsymbol{Ax}$, and it forms a 2-dimensional plane in 3-dimensional space $\mathbb{R}^3$.

Obviously, not all 3D vectors belong to range($\boldsymbol{A}$), so the linear system $\boldsymbol{Ax} = \boldsymbol{b}$ may not have a solution: it is **over-determined**. We may instead speak of a **least squares solution**: minimise the error $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{Ax}$,

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{b} - \boldsymbol{Ax}\|^2 .$$

For more details, see the notes on least squares solutions — that is Lecture 11.

Example 3.2 (Is $\boldsymbol{b}$ in the range?)**. Given.** The matrix of Example 3.1, $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.
Find. (a) Does $\boldsymbol{Ax} = \boldsymbol{b}$ have a solution for $\boldsymbol{b} = (1, 2, 2)$? (b) For $\boldsymbol{b} = (1, 2, 3)$? (c) Which vectors $\boldsymbol{b}$ lie in range($\boldsymbol{A}$)?

Answer. Every vector in the range has the form

$$\boldsymbol{Ax} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ x_2 \\ x_2 \end{bmatrix}.$$

(a) Yes. The second and third rows both ask for $x_2 = 2$, and then the first row gives $x_1 = 1+2 = 3$. So $\boldsymbol{b} = 3\boldsymbol{a}_1 + 2\boldsymbol{a}_2$.

(b) No. The second and third rows ask for $x_2 = 2$ and $x_2 = 3$ at the same time. Both systems are over-determined, three equations in two unknowns; this one has no solution, because $\boldsymbol{b}$ is not in the range.

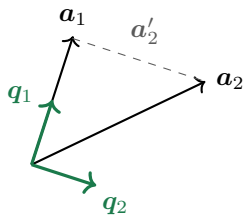
(c) Exactly the vectors with equal second and third entries: any (b_1, b_2, b_2) equals $\boldsymbol{Ax}$ with $x_2 = b_2$ and $x_1 = b_1 + b_2$. So range($\boldsymbol{A}$) is the plane $b_2 = b_3$. The vector $\boldsymbol{n} = (0, 1, -1)$ is orthogonal to both columns, $\boldsymbol{n}^\mathsf{T}\boldsymbol{a}_1 = 0$ and $\boldsymbol{n}^\mathsf{T}\boldsymbol{a}_2 = 0 + 1 - 1 = 0$, and the test $b_2 = b_3$ is the same as $\boldsymbol{n}^\mathsf{T}\boldsymbol{b} = 0$. For $\boldsymbol{b} = (1, 2, 3)$, $\boldsymbol{n}^\mathsf{T}\boldsymbol{b} = -1$.

Example 3.3 (Dependent columns)**. Given.** $\boldsymbol{A} = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{smallmatrix}\right)$.
Find. Does $\boldsymbol{A}$ have full column rank? What is range($\boldsymbol{A}$)?

Answer. No. The second column is twice the first, $\boldsymbol{a}_2 = 2\boldsymbol{a}_1$, so the columns are dependent and the rank is 1. Every $\boldsymbol{Ax} = x_1\boldsymbol{a}_1 + 2x_2\boldsymbol{a}_1 = (x_1 + 2x_2)\,\boldsymbol{a}_1$ is a multiple of $\boldsymbol{a}_1 = (1, 2, 3)$: the range is a line in $\mathbb{R}^3$, not a plane. The two columns are not a basis of the range; $\boldsymbol{a}_1$ alone is. Example 4.4 shows what Gram–Schmidt does with columns like these.

4 The Gram–Schmidt orthogonalisation process

Basic idea: the columns of $\boldsymbol{A}$ form the edges of a parallelepiped. "Straighten up" this parallelepiped into a cube, one edge at a time.



An orthonormal basis for Range($\boldsymbol{A}$) is built up as follows.

Step #	Orthogonalisation	Normalisation
1	N/A	$\boldsymbol{q}_1 = \boldsymbol{a}_1 / \|\boldsymbol{a}_1\|$
2	$\boldsymbol{a}_2' = \boldsymbol{a}_2 - (\boldsymbol{q}_1^\mathsf{T}\boldsymbol{a}_2)\,\boldsymbol{q}_1$	$\boldsymbol{q}_2 = \boldsymbol{a}_2' / \|\boldsymbol{a}_2'\|$
3	$\boldsymbol{a}_3' = \boldsymbol{a}_3 - (\boldsymbol{q}_1^\mathsf{T}\boldsymbol{a}_3)\,\boldsymbol{q}_1 - (\boldsymbol{q}_2^\mathsf{T}\boldsymbol{a}_3)\,\boldsymbol{q}_2$	$\boldsymbol{q}_3 = \boldsymbol{a}_3' / \|\boldsymbol{a}_3'\|$
⋮	⋮	⋮
n	$\boldsymbol{a}_n' = \boldsymbol{a}_n - \sum_{j=1}^{n-1} (\boldsymbol{q}_j^\mathsf{T}\boldsymbol{a}_n)\,\boldsymbol{q}_j$	$\boldsymbol{q}_n = \boldsymbol{a}_n' / \|\boldsymbol{a}_n'\|$

Key idea

Every row of that table is the same move: subtract off everything the new column already has in common with the directions already fixed, then rescale what is left to length one. The quantity $(\boldsymbol{q}_j^\mathsf{T}\boldsymbol{a}_n)\,\boldsymbol{q}_j$ is the part of $\boldsymbol{a}_n$ lying along $\boldsymbol{q}_j$; removing it is what makes $\boldsymbol{a}_n'$ orthogonal to $\boldsymbol{q}_j$.

Why this needs independent columns

If $\boldsymbol{a}_k$ were a combination of $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_{k-1}$, then after subtracting off all of its components along $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k-1}$ nothing would be left: $\boldsymbol{a}_k' = \boldsymbol{0}$, and the normalisation step would divide by zero. That is exactly the full column rank hypothesis we made above, showing up as a division.

Example 4.1 (Gram–Schmidt in the plane)**. Given.** The columns of $\boldsymbol{A} = \left(\begin{smallmatrix} 3 & -2 \\ 4 & 4 \end{smallmatrix}\right)$, that is $\boldsymbol{a}_1 = (3, 4)$ and $\boldsymbol{a}_2 = (-2, 4)$.
Find. (a) The orthonormal basis $\boldsymbol{q}_1, \boldsymbol{q}_2$ that Gram–Schmidt produces. (b) Is it really orthonormal? (c) What changes if the two columns are taken in the other order?

Answer.

(a) *Step 1.* $\|\boldsymbol{a}_1\| = \sqrt{9+16} = 5$, so $\boldsymbol{q}_1 = \boldsymbol{a}_1/5 = (\frac{3}{5}, \frac{4}{5})$.

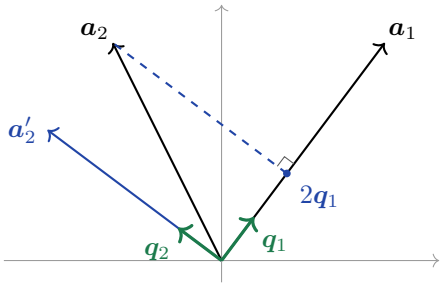
Step 2. The coefficient of $\boldsymbol{a}_2$ along $\boldsymbol{q}_1$ is $\boldsymbol{q}_1^\mathsf{T}\boldsymbol{a}_2 = (-6+16)/5 = 2$. Remove that part:

$$\boldsymbol{a}_2' = \boldsymbol{a}_2 - 2\,\boldsymbol{q}_1 = (-2, 4) - (\tfrac{6}{5}, \tfrac{8}{5}) = (-\tfrac{16}{5}, \tfrac{12}{5}).$$

Its length is $\|\boldsymbol{a}_2'\| = \frac{1}{5}\sqrt{256+144} = 4$, so $\boldsymbol{q}_2 = \boldsymbol{a}_2'/4 = (-\frac{4}{5}, \frac{3}{5})$.

(b) Yes. $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = (-12+12)/25 = 0$, and both have length 1 because $9+16=25$. Also $\boldsymbol{a}_1^{\mathsf{T}}\boldsymbol{a}_2' = (-48+48)/5 = 0$: removing the part along $\boldsymbol{q}_1$ is what made $\boldsymbol{a}_2'$ perpendicular to $\boldsymbol{a}_1$. The figure below shows the construction.

(c) Start from $\boldsymbol{a}_2$ instead. Then $\boldsymbol{q}_1 = \boldsymbol{a}_2/\|\boldsymbol{a}_2\| = (-1,2)/\sqrt{5}$, the part of $\boldsymbol{a}_1$ along it is $(\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_1)\,\boldsymbol{q}_1 = \sqrt{5}\,\boldsymbol{q}_1 = (-1,2)$, and what is left, $(3,4)-(-1,2) = (4,2)$, has length $2\sqrt{5}$, so $\boldsymbol{q}_2 = (2,1)/\sqrt{5}$. This is a different orthonormal basis of $\mathbb{R}^2$, and it is just as good. In either order the first $\boldsymbol{q}$ points along the first vector taken. In general Gram–Schmidt keeps the direction of the first vector, and $\boldsymbol{q}_1,\dots,\boldsymbol{q}_k$ span the same space as the first k vectors it was given.



Example 4.1. The dashed segment runs from $2\boldsymbol{q}_1$, the part of $\boldsymbol{a}_2$ along $\boldsymbol{q}_1$, to $\boldsymbol{a}_2$; it is $\boldsymbol{a}_2'$. Moved to the origin, $\boldsymbol{a}_2'$ points along $\boldsymbol{q}_2$.

Example 4.2 (An orthonormal basis for the plane of Example 3.1)**. Given.** The columns $\boldsymbol{a}_1 = (1,0,0)$ and $\boldsymbol{a}_2 = (-1,1,1)$ of the 3×2 matrix of Example 3.1.
Find. An orthonormal basis for range($\boldsymbol{A}$).

Answer. *Step 1.* $\|\boldsymbol{a}_1\| = 1$, so $\boldsymbol{q}_1 = \boldsymbol{a}_1 = (1,0,0)$. *Step 2.* $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = -1$. The coefficient is negative, so subtracting $(-1)\,\boldsymbol{q}_1$ means adding $\boldsymbol{q}_1$:

$$\boldsymbol{a}_2' = \boldsymbol{a}_2 - (-1)\,\boldsymbol{q}_1 = (-1,1,1)+(1,0,0) = (0,1,1), \qquad \|\boldsymbol{a}_2'\| = \sqrt{2}, \qquad \boldsymbol{q}_2 = \tfrac{1}{\sqrt{2}}\,(0,1,1).$$

Check: $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = 0$ and both have length 1. The plane in the figure of Section 3 now has a basis of two perpendicular unit vectors, and both have equal second and third entries, as Example 3.2(c) says every vector in this plane must.

Example 4.3 (Three vectors in $\mathbb{R}^3$, step by step)**. Given.** The columns of

$$\boldsymbol{A} = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 2 & 8 & 7 \end{pmatrix}: \qquad \boldsymbol{a}_1 = (1,2,2), \quad \boldsymbol{a}_2 = (-2,2,8), \quad \boldsymbol{a}_3 = (2,1,7).$$

Find. The orthonormal basis $\boldsymbol{q}_1,\boldsymbol{q}_2,\boldsymbol{q}_3$ from Gram–Schmidt, with the numbers that fill in the table of this section.

Answer. *Step 1.* $\|\boldsymbol{a}_1\| = \sqrt{1+4+4} = 3$, so $\boldsymbol{q}_1 = \frac{1}{3}(1,2,2)$.
Step 2. $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = \frac{1}{3}(-2+4+16) = 6$, so the part of $\boldsymbol{a}_2$ along $\boldsymbol{q}_1$ is $6\,\boldsymbol{q}_1 = (2,4,4)$, and

$$\boldsymbol{a}_2' = \boldsymbol{a}_2 - 6\,\boldsymbol{q}_1 = (-2,2,8)-(2,4,4) = (-4,-2,4).$$

Its length is $\sqrt{16+4+16} = 6$, so $\boldsymbol{q}_2 = \frac{1}{6}(-4,-2,4) = \frac{1}{3}(-2,-1,2)$.
Step 3. There are two coefficients now, $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{3}(2+2+14) = 6$ and $\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{3}(-4-1+14) = 3$. Subtract both parts:

$$\boldsymbol{a}_3' = \boldsymbol{a}_3 - 6\,\boldsymbol{q}_1 - 3\,\boldsymbol{q}_2 = (2,1,7)-(2,4,4)-(-2,-1,2) = (2,-2,1).$$

Its length is 3, so $\boldsymbol{q}_3 = \frac{1}{3}(2,-2,1)$.
Check. $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_2 = \frac{1}{9}(-2-2+4) = 0$, $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{q}_3 = \frac{1}{9}(2-4+2) = 0$ and $\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{q}_3 = \frac{1}{9}(-4+2+2) = 0$, and each $\boldsymbol{q}_i$ has length 1. It is worth checking $\boldsymbol{a}_3'$ against the original columns as well: $\boldsymbol{a}_3' \cdot \boldsymbol{a}_1 = 2-4+2 = 0$ and $\boldsymbol{a}_3' \cdot \boldsymbol{a}_2 = -4-4+8 = 0$.
In the table of this section the run reads:

Step #	Orthogonalisation	Normalisation
1	N/A	$\boldsymbol{q}_1 = \boldsymbol{a}_1/3 = \frac{1}{3}(1,2,2)$
2	$\boldsymbol{a}_2' = \boldsymbol{a}_2 - 6\,\boldsymbol{q}_1 = (-4,-2,4)$	$\boldsymbol{q}_2 = \boldsymbol{a}_2'/6 = \frac{1}{3}(-2,-1,2)$
3	$\boldsymbol{a}_3' = \boldsymbol{a}_3 - 6\,\boldsymbol{q}_1 - 3\,\boldsymbol{q}_2 = (2,-2,1)$	$\boldsymbol{q}_3 = \boldsymbol{a}_3'/3 = \frac{1}{3}(2,-2,1)$

Postponing the square roots

By hand it is often easier to make the vectors orthogonal first and normalise only at the end. Use the unnormalised $\boldsymbol{a}_j'$ in place of the $\boldsymbol{q}_j$, and divide each coefficient by a length squared instead:

$$\boldsymbol{a}_k' = \boldsymbol{a}_k - \sum_{j<k} \frac{\boldsymbol{a}_j' \cdot \boldsymbol{a}_k}{\boldsymbol{a}_j' \cdot \boldsymbol{a}_j'}\, \boldsymbol{a}_j', \qquad \text{starting from } \boldsymbol{a}_1' = \boldsymbol{a}_1.$$

In Example 4.3 this gives $\boldsymbol{a}_2' = \boldsymbol{a}_2 - \frac{18}{9}\,\boldsymbol{a}_1 = (-4,-2,4)$ and $\boldsymbol{a}_3' = \boldsymbol{a}_3 - \frac{18}{9}\,\boldsymbol{a}_1 - \frac{18}{36}\,\boldsymbol{a}_2' = (2,-2,1)$, the same vectors as before. The usual slip is to forget the division by $\boldsymbol{a}_j' \cdot \boldsymbol{a}_j'$. These coefficients are not the entries of $\boldsymbol{R}$: multiply each by the length $\|\boldsymbol{a}_j'\|$, so $r_{12} = 3 \cdot 2 = 6$, $r_{13} = 3 \cdot 2 = 6$ and $r_{23} = 6 \cdot \frac{1}{2} = 3$.

Example 4.4 (Dependent columns: Gram–Schmidt stops)**. Given.** The columns $\boldsymbol{a}_1, \boldsymbol{a}_2$ of Example 4.3, with the third column replaced by their sum, $\boldsymbol{a}_3 = \boldsymbol{a}_1 + \boldsymbol{a}_2 = (-1,4,10)$.
Find. What happens in step 3?
Answer. Steps 1 and 2 are unchanged. In step 3, $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{3}(-1+8+20) = 9$ and $\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{3}(2-4+20) = 6$, so

$$\boldsymbol{a}_3' = \boldsymbol{a}_3 - 9\,\boldsymbol{q}_1 - 6\,\boldsymbol{q}_2 = (-1,4,10) - (3,6,6) - (-4,-2,4) = (0,0,0).$$

Nothing is left, and $\boldsymbol{q}_3 = \boldsymbol{a}_3'/\|\boldsymbol{a}_3'\|$ would be $\boldsymbol{0}/0$. This is the box "Why this needs independent columns" in numbers: $\boldsymbol{a}_3$ already lies in the plane of $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$, the matrix has rank 2, and its range is that plane. In that case we drop $\boldsymbol{a}_3$, and $\boldsymbol{q}_1, \boldsymbol{q}_2$ is an orthonormal basis of the range. For the matrix of Example 3.3, where $\boldsymbol{a}_2 = 2\boldsymbol{a}_1$, the same thing happens at step 2.

Example 4.5 (A tall matrix: four rows, three columns)**. Given.** The columns of

$$\boldsymbol{A} = \begin{pmatrix} 1 & 5 & 4 \\ 1 & 5 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 0 \end{pmatrix}: \qquad \boldsymbol{a}_1 = (1,1,1,1), \quad \boldsymbol{a}_2 = (5,5,1,1), \quad \boldsymbol{a}_3 = (4,2,2,0).$$

Find. An orthonormal basis for range($\boldsymbol{A}$), a three-dimensional subspace of $\mathbb{R}^4$.
Answer. The steps are the same as in $\mathbb{R}^3$; only the vectors are longer.
Step 1. $\|\boldsymbol{a}_1\| = \sqrt{4} = 2$, so $\boldsymbol{q}_1 = \frac{1}{2}(1,1,1,1)$.

Step 2. $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = \frac{1}{2}(5+5+1+1) = 6$, so $\boldsymbol{a}_2' = (5,5,1,1) - (3,3,3,3) = (2,2,-2,-2)$, which has length 4, and $\boldsymbol{q}_2 = \frac{1}{2}(1,1,-1,-1)$.

Step 3. $\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{2}(4+2+2+0) = 4$ and $\boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{a}_3 = \frac{1}{2}(4+2-2-0) = 2$, so $\boldsymbol{a}_3' = (4,2,2,0) - (2,2,2,2) - (1,1,-1,-1) = (1,-1,1,-1)$, which has length 2, and $\boldsymbol{q}_3 = \frac{1}{2}(1,-1,1,-1)$.

The sign patterns $(+,+,+,+)$, $(+,+,-,-)$ and $(+,-,+,-)$ make the orthogonality easy to see. A fourth unit vector, $\boldsymbol{q}_4 = \frac{1}{2}(1,-1,-1,1)$, is orthogonal to all three and would complete a basis of $\mathbb{R}^4$; it is not in range$(\boldsymbol{A})$.

5 The QR decomposition

Just like LU is a decomposition representing the elimination process, QR is a decomposition representing the orthonormalisation process:

$$\underbrace{\begin{pmatrix} | & & | \\ \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n \\ | & & | \end{pmatrix}}_{\boldsymbol{A}\in\mathbb{R}^{m\times n}} = \underbrace{\begin{pmatrix} | & & | \\ \boldsymbol{q}_1 & \cdots & \boldsymbol{q}_n \\ | & & | \end{pmatrix}}_{\boldsymbol{Q}\in\mathbb{R}^{m\times n}} \underbrace{\begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ & r_{22} & \cdots & r_{2n} \\ & & \ddots & \vdots \\ \boldsymbol{0} & & & r_{nn} \end{pmatrix}}_{\boldsymbol{R}\in\mathbb{R}^{n\times n}}.$$

Reading the product one column at a time,

$$\boldsymbol{a}_1 = r_{11}\boldsymbol{q}_1, \qquad \boldsymbol{a}_2 = r_{12}\boldsymbol{q}_1 + r_{22}\boldsymbol{q}_2, \qquad \ldots, \qquad \boldsymbol{a}_n = \sum_j r_{jn}\,\boldsymbol{q}_j.$$

$\boldsymbol{R}$ is upper triangular for a structural reason: $\boldsymbol{a}_k$ was built using only $\boldsymbol{q}_1,\ldots,\boldsymbol{q}_k$, so no later $\boldsymbol{q}$ can appear in its expansion.

Exercise 5.1. Can you see how the r_{ij} are extracted from the steps in the table of Section 4, the one just before Example 4.1?

Unwind the table. Step 1 says $\boldsymbol{q}_1 = \boldsymbol{a}_1/\|\boldsymbol{a}_1\|$, so $\boldsymbol{a}_1 = \|\boldsymbol{a}_1\|\,\boldsymbol{q}_1$ and $r_{11} = \|\boldsymbol{a}_1\|$. Step 2 says $\boldsymbol{a}_2' = \boldsymbol{a}_2 - (\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2)\boldsymbol{q}_1$ and $\boldsymbol{a}_2' = \|\boldsymbol{a}_2'\|\,\boldsymbol{q}_2$, so

$$\boldsymbol{a}_2 = (\boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2)\,\boldsymbol{q}_1 + \|\boldsymbol{a}_2'\|\,\boldsymbol{q}_2 \qquad \Longrightarrow \qquad r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2, \quad r_{22} = \|\boldsymbol{a}_2'\|\,.$$

The same reading of row n gives $r_{jn} = \boldsymbol{q}_j^{\mathsf{T}}\boldsymbol{a}_n$ for $j<n$ and $r_{nn} = \|\boldsymbol{a}_n'\|$. The off-diagonal entries of $\boldsymbol{R}$ are the projection coefficients we subtracted off; the diagonal entries are the lengths we divided by.

Example 5.2 (Reading $\boldsymbol{Q}$ and $\boldsymbol{R}$ off Gram–Schmidt)**. Given.** The Gram–Schmidt run of Example 4.3.
Find. (a) $\boldsymbol{Q}$ and $\boldsymbol{R}$ with $\boldsymbol{A} = \boldsymbol{QR}$. (b) Do the second and third columns of $\boldsymbol{QR}$ give back $\boldsymbol{a}_2$ and $\boldsymbol{a}_3$? (c) Is $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{A}$ equal to $\boldsymbol{R}$?
Answer.

(a) The $\boldsymbol{q}$'s are the columns of $\boldsymbol{Q}$. The diagonal of $\boldsymbol{R}$ holds the lengths we divided by, 3, 6, 3. Above the diagonal are the coefficients we subtracted, $r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = 6$, $r_{13} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_3 = 6$ and $r_{23} = \boldsymbol{q}_2^{\mathsf{T}}\boldsymbol{a}_3 = 3$:

$$\underbrace{\begin{pmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 2 & 8 & 7 \end{pmatrix}}_{\boldsymbol{A}} = \underbrace{\frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{pmatrix}}_{\boldsymbol{Q}} \underbrace{\begin{pmatrix} 3 & 6 & 6 \\ 0 & 6 & 3 \\ 0 & 0 & 3 \end{pmatrix}}_{\boldsymbol{R}}.$$

(b) Yes. Column 2 of $\boldsymbol{QR}$ is $6\,\boldsymbol{q}_1 + 6\,\boldsymbol{q}_2 = (2,4,4) + (-4,-2,4) = (-2,2,8) = \boldsymbol{a}_2$, and column 3 is $6\,\boldsymbol{q}_1 + 3\,\boldsymbol{q}_2 + 3\,\boldsymbol{q}_3 = (2,4,4) + (-2,-1,2) + (2,-2,1) = (2,1,7) = \boldsymbol{a}_3$.

(c) Yes. Multiplying $\boldsymbol{A} = \boldsymbol{QR}$ on the left by $\boldsymbol{Q}^{\mathsf{T}}$ and using $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}$ gives $\boldsymbol{R} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{A}$. In numbers, with entry (i,j) equal to $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_j$:

$$\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{A} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ -2 & -1 & 2 \\ 2 & -2 & 1 \end{pmatrix}\begin{pmatrix} 1 & -2 & 2 \\ 2 & 2 & 1 \\ 2 & 8 & 7 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 9 & 18 & 18 \\ 0 & 18 & 9 \\ 0 & 0 & 9 \end{pmatrix} = \boldsymbol{R}.$$

The zeros below the diagonal come from orthogonality: $\boldsymbol{a}_j$ is a combination of $\boldsymbol{q}_1, \dots, \boldsymbol{q}_j$, so $\boldsymbol{q}_i^{\mathsf{T}}\boldsymbol{a}_j = 0$ whenever $i > j$.

Example 5.3 (A tall $\boldsymbol{Q}$)**. Given.** The 3×2 matrix of Example 3.1 and its Gram–Schmidt run in Example 4.2.
Find. (a) $\boldsymbol{Q}$ and $\boldsymbol{R}$. (b) $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q}$ and $\boldsymbol{Q}\boldsymbol{Q}^{\mathsf{T}}$.
Answer.

(a) Here $\boldsymbol{Q}$ is 3×2 and $\boldsymbol{R}$ is 2×2, as in the display at the start of this section, with $r_{11} = 1$, $r_{12} = \boldsymbol{q}_1^{\mathsf{T}}\boldsymbol{a}_2 = -1$ and $r_{22} = \sqrt{2}$:

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 0 & \sqrt{2} \end{pmatrix}.$$

Column 2 checks: $-\boldsymbol{q}_1 + \sqrt{2}\,\boldsymbol{q}_2 = (-1,0,0) + (0,1,1) = (-1,1,1)$.

(b) $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}_2$, because the two columns are orthonormal. But

$$\boldsymbol{Q}\boldsymbol{Q}^{\mathsf{T}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \neq \boldsymbol{I}_3.$$

A tall $\boldsymbol{Q}$ has no inverse, since two columns cannot span $\mathbb{R}^3$; $\boldsymbol{Q}^{\mathsf{T}}$ undoes $\boldsymbol{Q}$ from the left only. The rule $\boldsymbol{Q}^{-1} = \boldsymbol{Q}^{\mathsf{T}}$ of Section 2 is for square $\boldsymbol{Q}$. The matrix $\boldsymbol{Q}\boldsymbol{Q}^{\mathsf{T}}$ comes back in Lecture 11.

Example 5.4 (Volume from the diagonal of $\boldsymbol{R}$)**. Given.** The square matrices of Examples 4.1 and 4.3.
Find. How does $|\det \boldsymbol{A}|$ compare with the product of the diagonal entries of $\boldsymbol{R}$?
Answer. They are equal. For a square $\boldsymbol{A} = \boldsymbol{QR}$, $\det \boldsymbol{A} = \det \boldsymbol{Q} \det \boldsymbol{R}$. The determinant of the triangular $\boldsymbol{R}$ is the product of its diagonal, and $\det \boldsymbol{Q} = \pm 1$, because $(\det \boldsymbol{Q})^2 = \det \boldsymbol{Q}^{\mathsf{T}} \det \boldsymbol{Q} = \det(\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q}) = \det \boldsymbol{I} = 1$. So $|\det \boldsymbol{A}| = |r_{11}\, r_{22} \cdots r_{nn}|$, and for the $\boldsymbol{R}$ of Gram–Schmidt, whose diagonal entries are positive lengths,

$$|\det \boldsymbol{A}| = r_{11}\, r_{22} \cdots r_{nn}.$$

In Example 4.1, $\det \boldsymbol{A} = 3 \cdot 4 - (-2) \cdot 4 = 20 = 5 \cdot 4$. In Example 4.3, $\det \boldsymbol{A} = 54 = 3 \cdot 6 \cdot 3$. This is the parallelepiped at the start of Section 4: $r_{11} = \|\boldsymbol{a}_1\|$ is the length of the first edge, $r_{22} = \|\boldsymbol{a}_2'\|$ is the height of the second edge above the line of the first, r_{33} is the height of the third above the plane of the first two, and the volume is the product. In the plane it is base times height, 5×4 in the figure of Example 4.1.

Checking on the computer

`numpy.linalg.qr` and MATLAB's `qr` compute $\boldsymbol{A} = \boldsymbol{QR}$ by another method (Householder reflections), and they may flip the signs of some columns of $\boldsymbol{Q}$ together with the matching rows of $\boldsymbol{R}$. For the matrix of Example 5.2, numpy returns $\boldsymbol{R}$ with diagonal $-3, -6, 3$ and the first two columns of $\boldsymbol{Q}$ negated. Both answers are correct, since flipping a column of $\boldsymbol{Q}$ and the matching row of $\boldsymbol{R}$ leaves the product unchanged. For independent columns, the factorisation with the $m \times n$ matrix $\boldsymbol{Q}$ of this section is unique once we ask for $r_{kk} > 0$, and that is the one Gram–Schmidt produces. For a tall $\boldsymbol{A}$, `numpy.linalg.qr` returns the $m \times n$ matrix $\boldsymbol{Q}$ of this section, while MATLAB's `qr(A)` returns a square $m \times m$ one; `qr(A,0)` gives the $m \times n$ one.

6 Applications of the QR decomposition

① Solving linear equations

Given a square $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, we have $\boldsymbol{Q} \in \mathbb{R}^{n \times n}$ an orthogonal matrix and $\boldsymbol{R} \in \mathbb{R}^{n \times n}$ upper triangular (like the $\boldsymbol{U}$ in LU). Then

$$\boldsymbol{Ax} = \boldsymbol{b} \implies \boldsymbol{QRx} = \boldsymbol{b} \implies \boldsymbol{Rx} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b},$$

which is solved easily by back-substitution over $\boldsymbol{R}$.

The middle step is the whole point: moving $\boldsymbol{Q}$ to the other side needs $\boldsymbol{Q}^{-1}$, and $\boldsymbol{Q}^{-1} = \boldsymbol{Q}^{\mathsf{T}}$ is free.

Example 6.1 (Solving a square system with $\boldsymbol{QR}$)**. Given.** $\boldsymbol{A} = \boldsymbol{QR}$ from Example 5.2, and $\boldsymbol{b} = (7, 2, 8)$.
Find. The solution of $\boldsymbol{Ax} = \boldsymbol{b}$.
Answer. First multiply by $\boldsymbol{Q}^{\mathsf{T}}$, which takes three dot products:

$$\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b} = \tfrac{1}{3}\,(7+4+16,\ -14-2+16,\ 14-4+8) = (9, 0, 6).$$

Then solve $\boldsymbol{Rx} = \boldsymbol{Q}^{\mathsf{T}}\boldsymbol{b}$ by back-substitution, from the bottom row up:

$$\begin{aligned} 3x_3 = 6 &\implies x_3 = 2, \\ 6x_2 + 3x_3 = 0 &\implies x_2 = -1, \\ 3x_1 + 6x_2 + 6x_3 = 9 &\implies x_1 = 1. \end{aligned}$$

Check: $\boldsymbol{Ax} = (1+2+4,\ 2-2+2,\ 2-8+14) = (7, 2, 8)$. As with $\boldsymbol{L}$ and $\boldsymbol{U}$, the factorisation is done once; each new right-hand side then costs one multiplication by $\boldsymbol{Q}^{\mathsf{T}}$ and one back-substitution.

② Solving least squares problems

See the separate notes on least squares problems — Lecture 11, where the over-determined system of Example 3.1 finally gets solved.

Where this is going

Next time: least squares. We have the rectangular $\boldsymbol{A}$, the picture of range($\boldsymbol{A}$) as a plane that $\boldsymbol{b}$ generally misses, and now an orthonormal basis for that plane. Putting them together turns $\min_{\boldsymbol{x}} \|\boldsymbol{b} - \boldsymbol{Ax}\|^2$ into a projection, and QR is the numerically sound way to compute it.

Reading

Strang, *Computational Science and Engineering*, § 1.7 (numerical linear algebra: LU, QR, SVD); § 2.3 for the least squares problem that motivates it.

Practice for Quiz 1

These four exercises review Lectures 1 to 9, the material of Quiz 1. Try each one before reading its answer; the answers start on page 15.

Exercise 1 (Finite differences and a point load)**. Given.** The one-sided difference

$$D_h u(x) = \frac{-3\,u(x) + 4\,u(x+h) - u(x+2h)}{2h},$$

and the fixed–fixed problem

$$-\frac{d^2u}{dx^2} = \delta(x - \tfrac{1}{3}), \qquad 0 \le x \le 1, \qquad u(0) = 0, \quad u(1) = 0.$$

Find. (a) Use Taylor series to write $D_h u(x) = u'(x) + C\,h^2\,u'''(x) + \mathcal{O}(h^3)$. What is C, and what is the order of accuracy? (b) Check (a) on $u = x^3$ at $x = 0$ with $h = 0.1$. (c) What is the solution $u(x)$ of the boundary value problem? (d) By how much does u' jump at $x = \frac{1}{3}$, and why?

Exercise 2 (Eigenvalues and a system of differential equations)**. Given.**

$$\frac{d\boldsymbol{x}}{dt} = -\boldsymbol{A}\boldsymbol{x}, \qquad \boldsymbol{A} = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}, \qquad \boldsymbol{x}(0) = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

Find. (a) The eigenvalues and orthonormal eigenvectors of $\boldsymbol{A}$. Is $\boldsymbol{A}$ positive definite? (b) What is $\boldsymbol{x}(t)$? (c) The second component x_2 starts at 0. When is it largest, and what is its largest value? (d) What does $\boldsymbol{x}(t)$ look like for large t?

Exercise 3 (The Hessian and the second derivative test)**. Given.** $f(x_1, x_2) = x_1^3 + 3x_1x_2^2 - 3x_1$. **Find.** (a) The gradient and the Hessian of f. (b) All the critical points. (c) Is each critical point a local minimum, a local maximum or a saddle? What is f there?

Exercise 4 (The heat equation on a rod of length 2)**. Given.** A rod $0 \le x \le 2$ with both ends held at zero:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \qquad u(0,t) = 0, \quad u(2,t) = 0, \qquad u(x,0) = 5\sin\frac{\pi x}{2} - \sin \pi x.$$

Find. (a) Which modes do these ends allow, and how fast does each decay? (b) What is $u(x,t)$? (c) What is the temperature at the midpoint $x = 1$, and when has it fallen to half its starting value? (d) Show that $u(x,t) > 0$ for $0 < x < 2$ and every $t \ge 0$.

Answers

Exercise 1.

(a) Expand both shifted values about x:

$$u(x+h) = u + h\,u' + \tfrac{1}{2}h^2u'' + \tfrac{1}{6}h^3u''' + \mathcal{O}(h^4),$$
$$u(x+2h) = u + 2h\,u' + 2h^2u'' + \tfrac{4}{3}h^3u''' + \mathcal{O}(h^4).$$

In $-3u + 4u(x+h) - u(x+2h)$ the terms in u cancel ($-3+4-1=0$), the terms in u'' cancel ($2h^2 - 2h^2 = 0$), the u' terms give $4h - 2h = 2h$, and the u''' terms give $\frac{2}{3}h^3 - \frac{4}{3}h^3 = -\frac{2}{3}h^3$. Dividing by $2h$,

$$D_h u(x) = u'(x) - \tfrac{1}{3}h^2\,u'''(x) + \mathcal{O}(h^3).$$

So $C = -\frac{1}{3}$, and the formula is second order: halving h divides the error by about 4.

(b) $u(0) = 0$, $u(0.1) = 0.001$, $u(0.2) = 0.008$, so $D_h u(0) = (0.004 - 0.008)/0.2 = -0.02$, while $u'(0) = 0$. The formula predicts $-\frac{1}{3}(0.1)^2 \cdot 6 = -0.02$, since $u''' = 6$. The agreement is exact here because the fourth and higher derivatives of x^3 are zero.

(c) On each side of $x = \frac{1}{3}$, $u'' = 0$, so u is linear: $u = ax$ on the left (so that $u(0) = 0$) and $u = b(1-x)$ on the right (so that $u(1) = 0$). The two pieces meet at $x = \frac{1}{3}$, so $\frac{a}{3} = \frac{2b}{3}$. Integrating $-u'' = \delta(x - \frac{1}{3})$ across $x = \frac{1}{3}$ shows that the slope drops by 1 there, so $-b - a = -1$. Together, $a = \frac{2}{3}$ and $b = \frac{1}{3}$:

$$u(x) = \begin{cases} \frac{2}{3}\,x, & 0 \le x \le \frac{1}{3}, \\ \frac{1}{3}\,(1-x), & \frac{1}{3} \le x \le 1. \end{cases}$$

This is the fixed–fixed Green's function $g(x, y)$ at $y = \frac{1}{3}$, with its peak $u(\frac{1}{3}) = \frac{2}{9}$ under the load.

(d) u' drops from $\frac{2}{3}$ to $-\frac{1}{3}$, a jump of -1. The integral of $-u''$ across $x = \frac{1}{3}$ is minus the jump in u', and the integral of δ is 1: a unit point load puts a corner in u, while u itself does not jump.

Exercise 2.

(a) $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (3-\lambda)^2 - 1 = 0$ gives $\lambda_1 = 2$ and $\lambda_2 = 4$, with

$$\boldsymbol{v}_1 = \tfrac{1}{\sqrt{2}}\,(1,1), \qquad \boldsymbol{v}_2 = \tfrac{1}{\sqrt{2}}\,(1,-1).$$

Both eigenvalues are positive, so $\boldsymbol{A}$ is positive definite (the pivots, 3 and $\frac{8}{3}$, are positive too).

(b) Write $\boldsymbol{x}(0) = (4,0) = 2\,(1,1) + 2\,(1,-1)$. Each eigenvector part decays at its own rate:

$$\boldsymbol{x}(t) = 2\,e^{-2t}\begin{bmatrix}1\\1\end{bmatrix} + 2\,e^{-4t}\begin{bmatrix}1\\-1\end{bmatrix}, \qquad x_1 = 2e^{-2t} + 2e^{-4t}, \quad x_2 = 2e^{-2t} - 2e^{-4t}.$$

Check: at $t = 0$ this is $(4,0)$.

(c) $dx_2/dt = -4e^{-2t} + 8e^{-4t}$ is zero when $e^{-2t} = \frac{1}{2}$, that is at $t = \frac{1}{2}\ln 2 \approx 0.35$. There $x_2 = 2 \cdot \frac{1}{2} - 2 \cdot \frac{1}{4} = \frac{1}{2}$.

(d) The e^{-4t} part dies out first, so $\boldsymbol{x}(t) \approx 2e^{-2t}\,(1,1)$: the two components become equal and decay together at the rate of the smaller eigenvalue, 2.

Exercise 3.

(a)
$$\nabla f = \begin{bmatrix} 3x_1^2 + 3x_2^2 - 3 \\ 6x_1x_2 \end{bmatrix}, \qquad \boldsymbol{H} = \begin{pmatrix} 6x_1 & 6x_2 \\ 6x_2 & 6x_1 \end{pmatrix}.$$

(b) $6x_1x_2 = 0$ means $x_1 = 0$ or $x_2 = 0$. With $x_2 = 0$, $3x_1^2 = 3$ gives $x_1 = \pm 1$; with $x_1 = 0$, $3x_2^2 = 3$ gives $x_2 = \pm 1$. There are four critical points: $(1, 0)$, $(-1, 0)$, $(0, 1)$ and $(0, -1)$.

(c) At $(1, 0)$, $\boldsymbol{H} = 6\boldsymbol{I}$ is positive definite: a local minimum, with $f = -2$. At $(-1, 0)$, $\boldsymbol{H} = -6\boldsymbol{I}$ is negative definite: a local maximum, with $f = 2$. At $(0, \pm 1)$, $\boldsymbol{H} = \left(\begin{smallmatrix} 0 & \pm 6 \\ \pm 6 & 0 \end{smallmatrix}\right)$ has eigenvalues 6 and -6: two saddles, with $f = 0$. The minimum and maximum are only local: $f \to -\infty$ as $x_1 \to -\infty$ and $f \to +\infty$ as $x_1 \to +\infty$.

Exercise 4.

(a) Separating, $u = e^{-\lambda t} y(x)$ with $-y'' = \lambda y$ and $y(0) = y(2) = 0$. The solutions are
$$y_n(x) = \sin \frac{n\pi x}{2}, \qquad \lambda_n = \frac{n^2\pi^2}{4}, \qquad n = 1, 2, 3, \ldots$$
These are the modes of Lecture 9, with the length 2 in place of π.

(b) The starting temperature is already a sum of the modes $n = 1$ and $n = 2$, with coefficients 5 and -1, so
$$u(x, t) = 5\, e^{-\pi^2 t/4} \sin \frac{\pi x}{2} - e^{-\pi^2 t} \sin \pi x.$$

(c) Since $\sin \pi = 0$, the second mode is zero at the midpoint, and $u(1, t) = 5\, e^{-\pi^2 t/4}$. It starts at 5 and halves when $e^{-\pi^2 t/4} = \frac{1}{2}$, at $t = 4 \ln 2/\pi^2 \approx 0.28$.

(d) Use $\sin \pi x = 2 \sin \frac{\pi x}{2} \cos \frac{\pi x}{2}$ to take out the first mode:
$$u(x, t) = e^{-\pi^2 t/4} \sin \frac{\pi x}{2} \left(5 - 2\, e^{-3\pi^2 t/4} \cos \frac{\pi x}{2}\right).$$
For $0 < x < 2$ the sine is positive, and the bracket is at least $5 - 2 = 3$. So $u > 0$: the second mode starts five times smaller and decays four times faster, and it never outweighs the first.